

ANALYSIS OF URINE AND OTHER BODY FLUIDS

Laboratory Activity No. 1
Urine Specimen Collection and Handling

Processing Questions:

- A. **Sample Integrity Challenge:** A urine specimen arrives at the lab 4 hours after collection without refrigeration.

Question: Analyze how delayed transport and lack of proper storage could affect the accuracy of urinalysis results. What corrective actions should the medical technologist take?

- Delayed transport and lack of proper storage could change the urine composition and produce false-positive, false-negative, or unreliable urinalysis results. The specimen should reach the laboratory and be tested within 2 hours of collection; if delayed, it should be refrigerated at 4°C (yellow tube) or be at room temperature if using a gray tube.
- The medical technologist should reject the urine specimen and request recollection.

- B. **Type of Sample Dilemma:** A physician requests a midstream clean-catch sample, but the patient submits a random urine specimen instead.

Question: Evaluate the implications of using the wrong type of urine sample for diagnostic testing. Should the medical technologist proceed with analysis or request recollection?

- Using the wrong type of urine sample for diagnostic testing will lead to inaccurate results, misinterpretations, delayed diagnosis, or inappropriate treatment. The difference between midstream clean-catch samples and random urine specimens is that the former is designed to reduce contamination, while the latter is collected at any convenient time without requiring a particular timed collection.
- The medical technologist should not proceed with routine analysis. Request recollection unless the specimen is irreplaceable (e.g. ECF, tissue biopsies, etc.), which is not the case in the statement.

- C. **Contamination Scenario:** During processing, a medical technologist notices visible fecal contamination in a urine sample.

Question: Assess the potential impact of contamination on test results. What steps should the medical technologist follow to ensure patient safety and reliable reporting?

- Fecal contamination makes the urine specimen unreliable to diagnose, and this makes it a reason for rejection as this could form an incorrect diagnosis, although

considering whether the specimen is irreplaceable, the final decision should follow the laboratory's SOP.

- To ensure patient safety and reliable reporting, the medical technologist should stop routine reporting, assess the contaminated specimen, and request recollection whenever possible.

D. Specimen Labeling Issue: Two urine samples arrive without patient identifiers, and the ward staff insists they belong to different patients.

Question: Analyze the risks of processing unlabeled specimens. What is the most appropriate course of action for the medical technologist in this situation?

- Processing unlabeled specimens creates serious risks to patient identification, diagnostic accuracy, their treatment, and laboratory safety.
- The medical technologist should not analyze or report results until the identity can be confidently confirmed through an approved laboratory procedure, the most appropriate course of action is to reject the unlabeled urine specimen and request recollection.

E. Diagnostic Reasoning Case: A 24-hour urine collection is submitted, but the total volume is significantly less than expected, suggesting incomplete collection.

Question: Evaluate how incomplete collection affects quantitative tests (e.g., creatinine clearance, protein estimation). What should the medical technologist recommend to ensure valid results?

- An incomplete 24-hour urine collection makes false quantitative results because if part of the urine is missed, the measured volume is too low and the calculated daily excretion is underestimated and the total excretion is based on the urine collected over the entire 24-hour period.
- Urinary Creatinine excretion will be falsely low. This can produce inaccurate low clearance estimates and make renal filtration worse than it truly is.
- Total protein excretion will be underestimated. Patients with significant proteinuria could be incorrectly classified as having mild protein loss or no clinically important proteinuria.
- The medical technologist should request repeating the entire 24-hour urine collection, and to clarify to the patient that a low volume of urine suggests that it cannot provide reliable results and can lead to wrong diagnosis.